

Exercise 2: Implementing the Factory Method Pattern

```
interface Document {
```

```
    void open();
```

```
}
```

```
class WordDocument implements Document {
```

```
    public void open() {
```

```
        System.out.println("Opening Word Document");
```

```
    }
```

```
}
```

```
class PdfDocument implements Document {
```

```
    public void open() {
```

```
        System.out.println("Opening PDF Document");
```

```
    }
```

```
}
```

```
class ExcelDocument implements Document {
```

```
    public void open() {
```

```
        System.out.println("Opening Excel Document");
```

```
    }
```

```
}
```

```
abstract class DocumentFactory {
```

```
    abstract Document createDocument();  
}
```

```
class WordFactory extends DocumentFactory {  
    Document createDocument() {  
        return new WordDocument();  
    }  
}
```

```
class PdfFactory extends DocumentFactory {  
    Document createDocument() {  
        return new PdfDocument();  
    }  
}
```

```
class ExcelFactory extends DocumentFactory {  
    Document createDocument() {  
        return new ExcelDocument();  
    }  
}
```

```
public class FactoryMethodPattern {  
  
    public static void main(String[] args) {
```

DocumentFactory factory;

factory = new WordFactory();

Document doc1 = factory.createDocument();

doc1.open();

factory = new PdfFactory();

Document doc2 = factory.createDocument();

doc2.open();

factory = new ExcelFactory();

Document doc3 = factory.createDocument();

doc3.open();

}

}

FileEditSelectionViewGoRunTerminalHelp

CTS-JAVA-FSE-Exercises

Logger.java UFactoryMethodPattern.java U

Week1 > EngineeringConcepts > DesignPatterns > FactoryMethodPattern.java

```
DocumentFactory factory;

factory = new WordFactory();
Document doc1 = factory.createDocument();
doc1.open();

factory = new PdfFactory();
Document doc2 = factory.createDocument();
doc2.open();

factory = new ExcelFactory();
Document doc3 = factory.createDocument();
doc3.open();
}
```

PORTSTERMINALOUTPUTDEBUG CONSOLE

bash - DesignPatterns

```
$ javac FactoryMethodPattern.java
Tharun@LAPTOP-1791NH27 MINGW64 /f/coding/CTS-JAVA-FSE-Exercises/Week1/EngineeringConcepts/DesignPatterns (main)
$ java FactoryMethodPattern
Opening Word Document
Opening PDF Document
Opening Excel Document
Tharun@LAPTOP-1791NH27 MINGW64 /f/coding/CTS-JAVA-FSE-Exercises/Week1/EngineeringConcepts/DesignPatterns (main)
$
```

EXPLORER

CTS-JAVA-FSE-EXERCISES

Week1

EngineeringConcepts

DesignPatterns

Document.class U

DocumentFactory.class U

ExcelDocument.class U

ExcelFactory.class U

FactoryMethodPattern.class U

FactoryMethodPattern.java U

Logger.class U

Logger.java U

PdfDocument.class U

PdfFactory.class U

WordDocument.class U

WordFactory.class U

TIMELINE

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